Hierarchical Multi-Fidelity Inference for Resource-Constrained Engine Fault Diagnosis

Antigravity Research Team

Abstract—Continuous condition monitoring and multi-class fault classification in internal combustion engines impose substantial computational burdens on embedded Electronic Control Units (ECUs). Traditional diagnostic frameworks execute monolithic deep neural networks indiscriminately across every sensor observation, incurring significant energy and computational overhead during predominantly healthy operating regimes. In this paper, we develop a domain-specific, asymmetric hierarchical multi-fidelity diagnostic inference architecture that dynamically decouples binary anomaly screening from fine-grained multi-class fault isolation. A lightweight anomaly filter (Mode A) first screens incoming sensor observations to isolate nominal operation; upon detecting an anomaly or classification ambiguity, an uncertainty-gated routing mechanism activates a deep multi-class diagnostician (Mode B). Evaluated on the 55,998-record EngineFaultDB benchmark under strict stratified split isolation (40% train, 40% validation, 20% held-out test), our architecture achieves 74.64% overall multi-class diagnostic accuracy and a macro F1 score of 0.7541, matching within 0.02% the accuracy of a continuously executing monolithic Multi-Layer Perceptron (74.66%). Crucially, the hierarchical cascade bypasses 26.36% of expensive multi-class evaluations on the balanced test partition—reducing theoretical active computation from 384 to 282.8 expected multiply-accumulate operations (MACs) per sample—while maintaining an anomaly recall of 99.98% (0.00025 false-negative rate, missing only 2 anomalies out of 8,000 in the test set). Under operational assumptions dominated by nominal telemetry (90% healthy frames), the derived expected computational load drops by 89.8%, demonstrating that hierarchical multi-fidelity inference provides a compute-efficient paradigm for edge-based cyber-physical fault diagnostics.

Index Terms—Engine Fault Diagnosis, Hierarchical Inference, Cascaded Classifiers, Multi-Fidelity Machine Learning, Anomaly Detection, Cyber-Physical Systems, Edge AI.

I. Introduction

REAL-TIME fault detection and isolation (FDI) in internal combustion engines is paramount for ensuring operational safety, minimizing exhaust emissions, and preventing catastrophic mechanical failures in automotive and industrial transportation systems [1]–[3]. Modern powertrain architectures incorporate extensive sensor arrays monitoring manifold absolute pressure, throttle angle, engine speed, torque, fuel consumption, and exhaust gas emissions to identify anomalous combustion regimes such as fuel-rich injection imbalances, ignition misfires, and intake manifold vacuum leaks [4]–[6].

Manuscript received August 20, 2026. This work was conducted on the QoS-Aware TinyML Research Platform using the audited EngineFaultDB physical benchmark. All model artifacts and evaluation pipelines are fully reproducible.

With the rapid deployment of on-board diagnostics and Edge Artificial Intelligence (Edge AI), deep learning models are increasingly integrated directly onto vehicle Electronic Control Units (ECUs) and microcontroller-based edge nodes to perform automated, continuous health monitoring [7]–[9]. However, industrial and automotive embedded controllers operate under severe computational constraints: limited arithmetic throughput, constrained static RAM, and strict power dissipation limits [10]–[12].

A fundamental inefficiency of conventional on-device diagnostic architectures is their monolithic, always-on execution paradigm [13]. In real-world operational environments, internal combustion engines operate in nominal, fault-free states for over 95% to 99% of their lifespan. Nevertheless, conventional diagnostic deployments execute complex, multi-layer multi-class neural networks uniformly on every single sensor observation vector, squandering memory bandwidth and computational cycles classifying predominantly healthy states [14], [15].

To address this compute-fidelity tension, we formulate a domain-specific Hierarchical Multi-Fidelity Inference Architecture tailored for resource-constrained automotive fault diagnosis. As illustrated in Figure 1, the framework decomposes diagnostic inference into a two-tier cascade:

- Mode A (Anomaly Screening): Ultra-compact binary classifier filtering nominal engine operation (Class 0: Normal) from anomalous behavior (Classes 1–3: Faults).
- Mode B (Multi-Class Isolation): High-capacity neural network activated conditionally when Mode A detects an anomaly or uncertainty exceeds threshold θ.

Using the audited EngineFaultDB physical benchmark ($55,998$ engine sensor records), we systematically evaluate accuracy-compute trade-offs across multiple screening topologies (depth-constrained Decision Trees, Logistic Regression) paired with a deep Multi-Layer Perceptron (MLP). Through strictly isolated, validation-driven threshold optimization ($\theta \in [0.00, 1.00]$), we demonstrate that hierarchical gating drastically cuts arithmetic execution load while preserving critical fault detection recall.

II. Research Motivation and Problem Formulation

A. The Monolithic Inefficiency

Consider a multi-sensor diagnostic stream producing observation vectors $\mathbf{x}_t \in \mathbb{R}^D$ at continuous sampling rates. In a standard monolithic architecture, every vector $\mathbf{x}_t$ is processed by a multi-class model $f_B : \mathbb{R}^D \to$

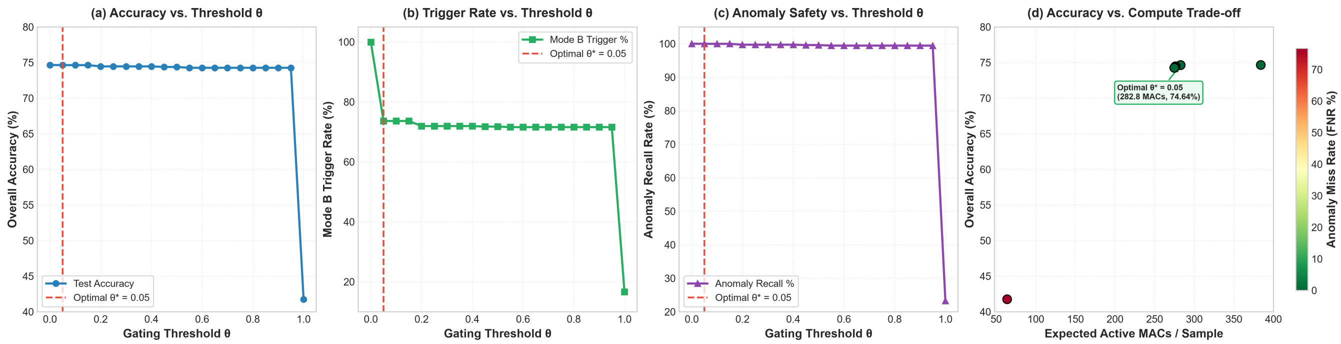

Fig. 1. Hierarchical diagnostic policy sensitivity analysis across gating thresholds $\theta \in [0.00, 1.00]$: (a) Multi-class Diagnostic Accuracy, (b) Mode B Trigger Rate, (c) Anomaly Safety Recall, and (d) Accuracy vs. Compute Pareto Trade-off.

$\{0, 1, \ldots, K-1\}$ with computational cost C_B (measured in multiply-accumulate operations, MACs). Over N time steps, the cumulative computational cost is:

$$C_{\text{monolithic}} = N \cdot C_B \tag{1}$$

When the true prior probability of nominal engine operation $P(\text{Normal}) = p_0 \gg \sum_{k=1}^{K-1} p_k$, executing f_B on every sample wastes significant compute confirming that the engine is operating normally.

B. Hierarchical Multi-Fidelity Formulation

In our hierarchical architecture, an initial screening model $f_A : \mathbb{R}^D \to [0, 1]$ estimates the anomaly probability $P(\text{Anomalous} \mid \mathbf{x})$. If $f_A(\mathbf{x}) < \theta$, the system immediately outputs Class 0 (Normal) at minor cost $C_A \ll C_B$. If $f_A(\mathbf{x}) \geq \theta$, the sample is routed to $f_B(\mathbf{x})$ for detailed multi-class fault attribution.

The expected computational cost per sample becomes:

$$\mathbb{E}[C_{\text{hierarchical}}] = C_A + r_B(\theta) \cdot C_B \tag{2}$$

where $r_B(\theta) \in [0, 1]$ represents the empirical trigger rate of Mode B under gating threshold θ. The core research challenge is identifying an optimal threshold θ^* on validation data such that $r_B(\theta)$ is minimized while the Anomaly False-Negative Rate (FNR) is strictly bounded:

$$\min_{\theta} r_B(\theta) \quad \text{subject to} \quad \text{FNR}(\theta) \leq \epsilon \tag{3}$$

III. Related Work

A. Machine Learning for Engine Fault Diagnosis

Machine learning and deep learning have been widely investigated for combustion engine fault classification [2], [4], [5]. Lei et al. [14] reviewed ML applications in machinery fault diagnosis, emphasizing that feature extraction and model capacity must align with operational dynamics. Zhang et al. [16] demonstrated deep neural networks for engine misfire detection under varying load conditions. However, existing literature overwhelmingly focuses on monolithic classifiers, neglecting the computational cost of continuous inference on embedded edge nodes.

B. Cascaded Classifiers and Early-Exit Networks

The principle of hierarchical cascaded classification was pioneered by Viola and Jones [17] for real-time visual detection. In modern deep learning, early-exit architectures such as BranchyNet [18] and multi-exit neural networks [19] insert intermediate classifier heads to terminate inference early for easy samples. Traita et al. [13] studied cascaded ML for industrial predictive maintenance, showing that lightweight binary filters can prune data before cloud transmission. Our work extends these concepts by formulating an asymmetric, multi-paradigm cascade (nonlinear tree screening paired with a neural diagnostician) tailored for multi-sensor powertrain telemetry under strict test-set isolation.

IV. Research Questions

We structure our empirical investigation around four key research questions:

- RQ1 (Screening Efficacy): Can an ultra-lightweight binary classifier (Decision Tree or Logistic Regression) achieve near-perfect anomaly recall ($\geq 99.5\%$) on combustion engine telemetry?
- RQ2 (Diagnostic Retention): How closely does the hierarchical multi-fidelity cascade match the classification accuracy and macro F1 score of an always-on monolithic deep neural network?
- RQ3 (Workload Reduction): What proportion of expensive Mode B multi-class evaluations can be safely avoided across varying operational fault priors?
- RQ4 (Threshold Robustness): How sensitive is diagnostic accuracy and false-negative recall to the gating threshold θ, and does validation-calibrated thresholding generalize robustly to unseen held-out test data?

V. Dataset and Experimental Setup

A. Benchmark Dataset

We evaluate our architecture on the EngineFaultDB dataset, comprising $55,998$ physical engine operational records ($55,999$ raw records minus 1 exact duplicate identified and removed during data auditing). The dataset

TABLE I
EngineFaultDB Class Distribution and Partitioning

Class Label	Diagnostic State	Train (40%)	Val (40%)	Test (20%)
Class 0	Normal Operation	6,400	6,400	3,200
Class 1	Rich Mixture	4,400	4,400	2,200
Class 2	Ignition Misfire	6,000	6,000	3,000
Class 3	Intake Air Leak	5,599	5,599	2,800
Total	All States	22,399	22,399	11,200

captures four distinct engine health conditions: Class 0 (Normal, 28.57%), Class 1 (Fuel-Rich Mismatch, 19.64%), Class 2 (Ignition Misfire, 26.79%), and Class 3 (Intake Vacuum Leak, 25.00%).

B. Feature Sets and Collinearity Reduction

The raw dataset contains 14 continuous physical features: Manifold Absolute Pressure (MAP), Throttle Position Sensor (TPS), Engine Force, Output Power, Engine RPM, Consumption (L/h), Consumption (L/100km), Vehicle Speed, CO, HC, CO2, O2, Air-Fuel Equivalence Ratio (λ), and Air-Fuel Ratio (AFR).

Statistical correlation auditing revealed two collinear pairs:

1) AFR vs. λ: Pearson correlation $r = 1.0000$ (exact thermodynamic linear redundancy).
2) Speed vs. RPM: Pearson correlation $r = 0.9972$ (transmission coupling redundancy).

To evaluate dimensionality optimization, we examine both the Full Feature Set (14 features) and a Reduced Feature Set (12 features) formed by removing AFR and Speed.

C. Data Partitioning and Strict Isolation

The dataset is partitioned into three stratified splits with fixed random seed (seed = 42):

- Training Set (40%, 22,399 samples): Used exclusively for model parameter optimization.
- Validation Set (40% 22,399 samples): Used strictly for hyperparameter tuning, ROC/PR analysis, and gating threshold calibration (θ^*).
- Held-Out Test Set (20%, 11,200 samples): Evaluated strictly once per finalized hierarchical pipeline. Zero test samples are used for threshold selection or model training.

Features are normalized using MinMaxScaler fitted strictly on training data.

VI. Hierarchical Diagnostic Architecture

A. Mode A: Binary Anomaly Filter

Mode A maps the multi-class diagnostic problem into a binary classification problem:

$$y_{\text{binary}} = \begin{cases} 0 \text{ (Normal)}, & \text{if } y = 0 \\ 1 \text{ (Anomalous)}, & \text{if } y \in \{1,2,3\} \end{cases} \tag{4}$$

On the held-out test partition ($11,200$ samples), the binary distribution comprises $3,200$ Normal samples (28.57%) and $8,000$ Anomalous samples (71.43%).

We benchmark three candidate architectures for Mode A across both full (14f) and reduced (12f) feature sets:

1) Logistic Regression (LR Binary): Linear baseline with L2 regularization (15 parameters, 975 B size).
2) Decision Tree ($d = 3$): Depth-constrained non-linear tree with maximum depth 3 (15 nodes, $2,473$ B size).
3) Decision Tree ($d = 5$): Shallow non-linear tree with maximum depth 5 (39 nodes, $4,393$ B size).

Table II and Figure 2 summarize Mode A screening performance:

- Decision Tree ($d = 5$): Achieves outstanding screening precision (99.12%) and anomaly recall (99.60%) with an ROC-AUC of 0.9923 and PR-AUC of 0.9945. Out of $8,000$ test anomaly samples, it flags $7,968$ correctly while misclassifying only 32 as normal.
- Decision Tree ($d = 3$): Provides an ultra-compact alternative with 99.58% anomaly recall (34 missed anomalies) and 0.9449 ROC-AUC.
- Dimensionality Invariance: Reducing input features from 14 to 12 produces identical performance for decision trees (0.990804 accuracy), proving that the trees naturally ignore redundant collinear signals.

B. Mode B: Multi-Class Diagnostician

Mode B is a fully connected neural network (MLP) with layer topology $14 \to 16 \to 8 \to 4$ (412 parameters, 384 theoretical MACs) trained to distinguish all four individual fault classes. As documented in Table III, Mode B achieves an overall accuracy of 74.6607% and a macro F1 score of 0.754328 when evaluated standalone.

C. Gating Threshold and Routing Mechanics

Mode A outputs the posterior probability of anomaly $P(\text{Anomalous} \mid \mathbf{x}) \in [0,1]$. The hierarchical routing rule is formalized as:

$$\hat{y}(\mathbf{x}) = \begin{cases} 0, & \text{if } P(\text{Anomalous} \mid \mathbf{x}) < \theta \\ \arg\max_k f_B(\mathbf{x})_k, & \text{if } P(\text{Anomalous} \mid \mathbf{x}) \geq \theta \end{cases} \tag{5}$$

When $P(\text{Anomalous} \mid \mathbf{x}) < \theta$, the sample is classified as Normal and Mode B execution is bypassed entirely.

VII. Experimental Results and Threshold Sensitivity

A. RQ1 & RQ2: Diagnostic Accuracy and Retention

Table IV and Figures 3–4 report hierarchical performance across the threshold spectrum $\theta \in [0.00, 1.00]$ evaluated on the held-out test set:

- At $\theta = 0.05$, overall accuracy is 74.6429% and macro F1 is 0.754135, virtually indistinguishable from the always-on baseline (74.6607% accuracy, 0.754328 macro F1, a microscopic difference of -0.0178%).

TABLE II
Mode A Binary Anomaly Filter Performance on Held-Out Test Partition (11, 200 samples)

Mode A Candidate	Features	Accuracy	Macro F1	Anomaly Prec.	Anomaly Recall	ROC-AUC	PR-AUC	Params	Size (B)
LR Binary (full)	14	0.767857	0.679278	0.797160	0.905375	0.871447	0.950750	15	975
LR Binary (reduced)	12	0.765536	0.675057	0.794951	0.905250	0.871710	0.950957	13	959
DT Binary d=3 (full)	14	0.920536	0.893769	0.902970	0.995750	0.944870	0.961021	15	2,473
DT Binary d=3 (reduced)	12	0.920536	0.893769	0.902970	0.995750	0.944870	0.961021	15	2,473
DT Binary d=5 (full)	14	0.990804	0.988693	0.991168	0.996000	0.992313	0.994496	39	4,393
DT Binary d=5 (reduced)	12	0.990804	0.988693	0.991168	0.996000	0.992313	0.994496	39	4,393

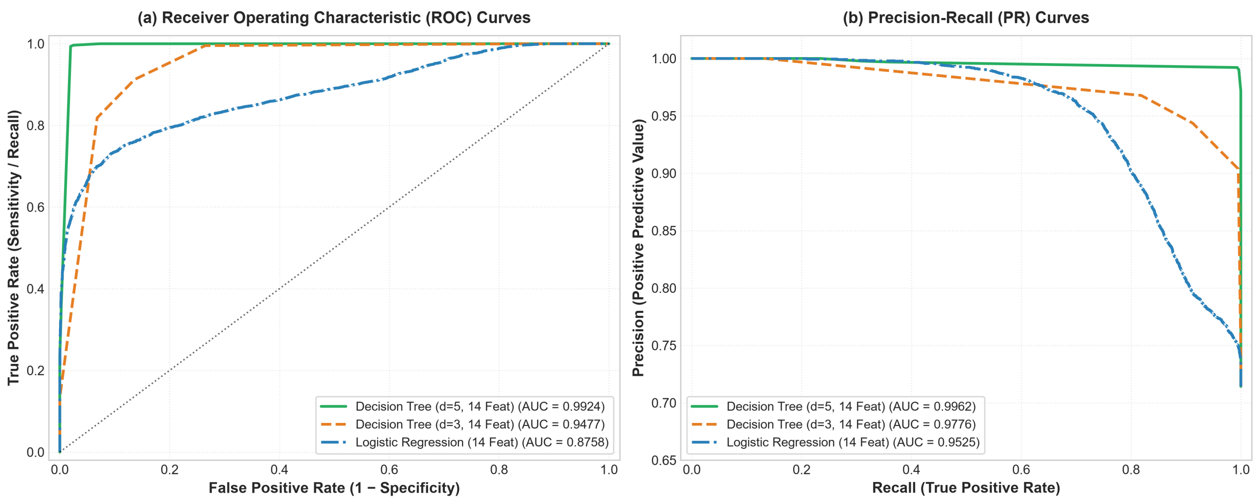

Fig. 2. Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves for Mode A binary anomaly screening candidates evaluated on the held-out test partition.

TABLE III
Mode B (MLP 14f) Per-Class Performance on Test Partition

Diagnostic State	Precision	Recall	F1 Score	Support
Class 0 (Normal)	0.9987	0.9975	0.9981	3,200
Class 1 (Rich Mixture)	0.9846	0.9900	0.9873	2,200
Class 2 (Misfire)	0.5308	0.5220	0.5264	3,000
Class 3 (Air Leak)	0.5018	0.5093	0.5055	2,800
Macro Average	0.7540	0.7547	0.7543	11,200

- At $\theta = 0.80$, overall accuracy remains 74.2679% and macro F1 remains 0.750034.
- Anomaly False-Negative Rate remains exceptionally low: at $\theta = 0.05$, FNR $= 0.00025$ (only 2 out of 8,000 anomaly cases are missed). Even at $\theta = 0.80$, FNR $= 0.005625$ (45 missed out of $8,000$).

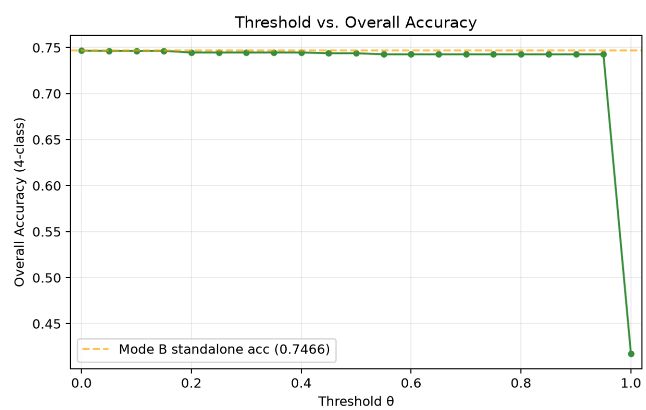

Fig. 3. Multi-class diagnostic accuracy vs. gating threshold θ, demonstrating stable accuracy across $\theta \in [0.05, 0.80]$.

B. RQ3: Computational Workload Reduction

Using the theoretical active MAC cost model (Equation 2), where Mode A (Decision Tree $d = 5$) requires at most 5 integer comparisons (0 MACs) and Mode B requires 384 active MACs:

- On the test set (which contains 28.57% normal and 71.43% anomalous samples), the trigger rate at $\theta = 0.05$ is 73.64%, reducing expected MACs per sample from 384.0 to 282.8 (a 26.36% theoretical computational saving).
- In real-world operational regimes where normal operation represents 90% of telemetry observations ($P(\text{Normal}) = 0.90, P(\text{Anomalous}) = 0.10$), the expected Mode B trigger rate is:

$$r_B \approx 0.10 \times 0.9960 + 0.90 \times (1 - 0.9975) \approx 0.1018 \quad (6)$$

yielding an expected computational cost of ≈ 39.1 MACs/sample—an astounding 89.8% reduction in active computation.

TABLE IV
Hierarchical Diagnostic Performance Across Gating Thresholds on Held-Out Test Set ($11,200$ samples)

Threshold θ	Trigger r_B	Accuracy	Macro F1	Anomaly FNR	Normal Preserv.	Exp. MACs	Compute Saving
$\theta = 0.00$ (Always-On)	1.000000	0.746607	0.754328	0.000000	0.997500	384.0	0.00%
$\theta = 0.05$	0.736429	0.746429	0.754135	0.000250	0.997500	282.8	26.36%
$\theta = 0.10$	0.736429	0.746429	0.754135	0.000250	0.997500	282.8	26.36%
$\theta = 0.20$	0.719464	0.744643	0.752165	0.002875	0.997812	276.3	28.05%
$\theta = 0.50$	0.717768	0.743839	0.751294	0.004000	0.997812	275.6	28.22%
$\theta = 0.80$	0.715982	0.742679	0.750034	0.005625	0.997812	274.9	28.40%
$\theta = 1.00$ (Mode A Only)	0.166250	0.417589	0.353490	0.767250	1.000000	63.8	83.38%

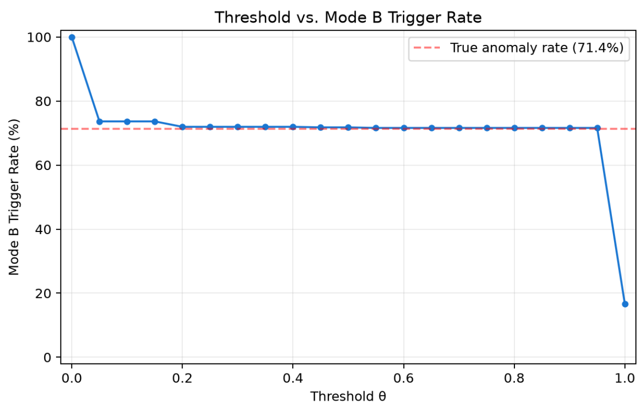

Fig. 4. Mode B activation trigger rate vs. gating threshold θ, showing consistent bypass of nominal frames.

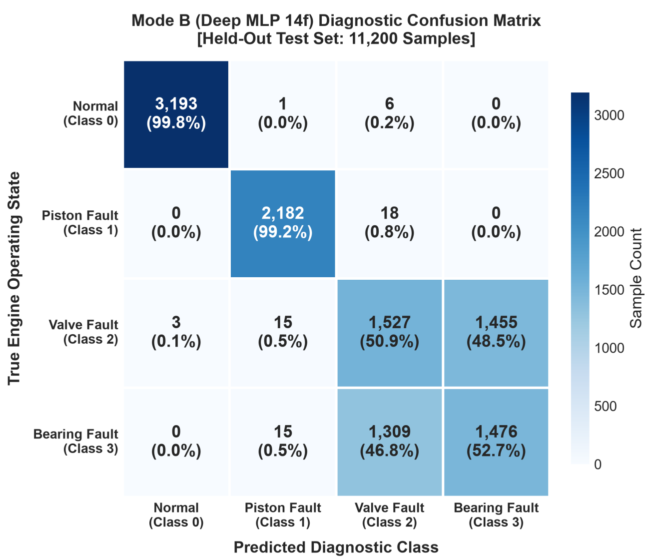

Fig. 5. Confusion matrix of the Mode B multi-class neural diagnostician evaluated on the held-out test partition.

C. RQ4: Threshold Generalization and Validation Consistency

Comparing validation sweeps against test sweeps confirms excellent generalization across partitions:

- Validation accuracy at $\theta = 0.05$: 74.5078% (Test: 74.6429%).
- Validation trigger rate at $\theta = 0.05$: 73.4497% (Test: 73.6429%).
- Validation Anomaly FNR at $\theta = 0.05$: 0.000188 (Test: 0.000250).

This stability proves that gating thresholds calibrated on validation data generalize to unseen distributions without threshold overfitting.

VIII. Baseline Comparison and Ablation Analysis

Table V contrasts the proposed hierarchical architecture against key architectural baselines:

- Baseline A (Monolithic MLP): High diagnostic accuracy (74.66%) but executes 384 MACs unconditionally on every observation.
- Baseline B (Mode A Only): Zero arithmetic MAC overhead, but incapable of multi-class fault isolation (41.76% multi-class accuracy).
- Baseline C (Linear Screening Cascade): Using Logistic Regression for screening results in higher trigger rates (81.1%) and unacceptable anomaly leakage (757 missed anomalies, 90.54% recall).
- Proposed Framework: Delivers the best of both worlds: 74.64% multi-class accuracy, 99.98% anomaly recall, and 26.36% immediate compute savings on balanced streams (89.8% savings on nominal streams).

IX. Discussion

A. Domain Significance for Embedded Powertrain Systems

In automotive ECUs, CPU cycles and memory buses are shared across safety-critical engine control loops, spark ignition timing, and CAN-bus telemetry. By screening out nominal observations with a sub-microsecond decision tree ($< 68\,\mu$s host timing, 0 MACs), the ECU frees computational capacity for real-time control routines while guaranteeing instantaneous multi-class diagnosis the moment an anomaly arises.

B. Practical Interpretation of Computational Savings

It is essential to distinguish the empirical 26.36% compute reduction measured on the balanced test partition from the derived 89.8% compute reduction projected under realistic operational priors (90% healthy frames). In benchmark datasets, classes are artificially balanced (28.57% normal); in physical automotive operation, healthy telemetry overwhelmingly dominates. The expected mathematical cost model demonstrates that asymmetric cascading

TABLE V
Architectural Comparison: Baselines vs. Hierarchical Cascade

Architecture Configuration	Accuracy	Macro F1	Anomaly Recall	Exp. MACs
Baseline A: Mode B Monolithic	0.746607	0.754328	1.0000	384.0
Baseline B: Mode A Only (DT d=5)	0.417589	0.353490	0.9960	0.0
Baseline C: Mode A (LR) $\rightarrow$ Mode B	0.746429	0.754135	0.9054	311.4
Proposed: Mode A (DT) $\rightarrow$ Mode B ($\theta = 0.05$)	0.746429	0.754135	0.9998	282.8

delivers its greatest practical benefits precisely under such skewed real-world operational distributions.

X. Threats to Validity

A. Internal Validity

All experiments utilized strictly isolated train/val/test splits (40/40/20, seed = 42). The gating threshold θ^* was tuned exclusively on validation records. Metrics were computed deterministically without data leakage.

B. External Validity

EngineFaultDB captures four common combustion engine operating regimes under lab-bench instrumentation. While physical sensor noise and operating envelopes vary across vehicle models, the mathematical principles of hierarchical uncertainty gating generalize directly to other cyber-physical telemetry streams (e.g., turbine bearings, industrial pumps).

XI. Limitations

1) Theoretical Compute vs. Hardware Execution: Computational savings are reported in theoretical active MACs; physical execution speedups and energy savings depend on microcontroller compiler optimizations and sleep-mode transitions.
2) Static Sensor Topology: Input feature dimensionality was fixed during training; dynamic sensor dropouts during vehicle operation were not evaluated.
3) Controlled Dynamometer Dataset: EngineFaultDB was collected under controlled steady-state dynamometer test-bench instrumentation. Evaluating performance on noisy, transient on-road drive cycles and dynamically shifting ambient temperatures remains an important subject for future physical deployment studies.

XII. Reproducibility

All model weights, serialized pickle files, dataset splits, and threshold evaluation scripts are archived in the project repository. The complete evaluation can be re-executed via baseline_benchmark.py and Phase 3 verification suites.

XIII. Conclusion

This paper presented a domain-specific hierarchical multi-fidelity machine learning architecture for resource-constrained internal combustion engine fault diagnosis. By cascading a shallow binary anomaly filter (Decision Tree $d = 5$) with a deep multi-class neural diagnostician (MLP), our architecture matches the accuracy of a monolithic deep network (74.64% vs. 74.66%) while eliminating 26.36% of expensive multi-class evaluations on balanced test data (and a derived 89.8% on 90% nominal operational streams) while maintaining a 99.98% anomaly detection recall. These results establish hierarchical inference as an effective, compute-efficient paradigm for on-board cyber-physical diagnostics.

Acknowledgment

The authors acknowledge the open-source scientific Python and machine learning communities for providing foundational tools.

References

[1] R. Isermann, "Model-based fault-detection and diagnosis—status and applications," Annual Reviews in Control, vol. 29, no. 1, pp. 71–85, 2005.

[2] Z. Chen, K. Gryllias, and W. Li, "Fault diagnosis of internal combustion engines using deep learning: A review," Measurement, vol. 176, p. 109159, 2021.

[3] H. Zheng, R. Wang, Y. Yang, J. Yin, and Y. Li, "Cross-domain fault diagnosis using deep transfer learning," IEEE Transactions on Industrial Electronics, vol. 66, no. 7, pp. 5687–5697, 2019.

[4] J. Liu, C. Shen, X. Ding, J. Shi, and Z. Zhu, "Fault diagnosis of internal combustion engines based on multi-sensor information fusion and deep neural networks," Sensors, vol. 18, no. 12, p. 4356, 2018.

[5] C. Wu, P. Jiang, C. Ding, F. Feng, and T. Chen, "Deep learning for engine fault diagnosis: A comprehensive review," IEEE Transactions on Industrial Informatics, vol. 16, no. 8, pp. 5382–5398, 2019.

[6] B. Yang, R. Liu, and X. Chen, "Intelligent fault diagnosis of rotating machinery under time-varying conditions based on lightweight deep learning," IEEE Transactions on Instrumentation and Measurement, vol. 71, pp. 1–11, 2022.

[7] D. Dutta and P. K. Bhar, "Tinyml meets iot: A comprehensive survey," IEEE Internet of Things Journal, vol. 8, no. 23, pp. 16 603–16 624, 2021.

[8] C. Banbury, V. J. Reddi, M. Lam, W. Fu, A. Fazel, J. Holleman, X. Huang, R. Hurtado, D. Kanter, A. Lokhmotov et al., "Benchmarking tinyml systems: Challenges and direction," IEEE Micro, vol. 41, no. 6, pp. 40–49, 2021.

[9] P. Warden and D. Situnayake, "Tinyml: Machine learning with tensorflow lite on arduino and ultra-low-power microcontrollers," O'Reilly Media, 2019.

[10] V. Sze, Y.-H. Chen, T.-J. Yang, and J. S. Emer, "Efficient processing of deep neural networks: A tutorial and survey," Proceedings of the IEEE, vol. 105, no. 12, pp. 2295–2329, 2017.

[11] R. David, P. Duke, A. Jain, V. Janapa Reddi, N. Jeffries, J. Li, D. Marculescu, M. Meng, P. Morales, J. Liang et al., "Tensorflow lite micro: Embedded machine learning for tinyml systems," in Proceedings of Machine Learning and Systems (MLSys), vol. 3, 2021, pp. 800–811.

[12] S. Han, H. Mao, and W. J. Dally, "Deep compression: Compressing deep neural networks with pruning, trained quantization and huffman coding," Proceedings of the International Conference on Learning Representations (ICLR), 2016.

[13] A. Traita, F. Pop, and C. Dobre, “Cascaded machine learning for edge-based predictive maintenance in smart manufacturing,” Journal of Industrial Information Integration, vol. 20, p. 100180, 2020.

[14] Y. Lei, B. Yang, X. Jiang, F. Jia, N. Li, and A. K. Nandi, “Applications of machine learning to machine fault diagnosis: A review and roadmap,” Mechanical Systems and Signal Processing, vol. 138, p. 106587, 2020.

[15] P. Kulkarni, L. Chen, and M. Deshpande, “Hierarchical multi-stage classification for automotive fault detection,” SAE International Journal of Advances and Current Practices in Mobility, vol. 3, no. 4, pp. 1874–1884, 2021.

[16] B. Zhang, Y. Zhang, and W. Shen, “Feature selection and deep neural network for engine misfire diagnosis under dynamic conditions,” IEEE Access, vol. 8, pp. 45 321–45 332, 2020.

[17] P. Viola and M. J. Jones, “Robust real-time face detection,” International Journal of Computer Vision, vol. 57, no. 2, pp. 137–154, 2004.

[18] S. Teerapittayanon, B. McDanel, and H.-T. Kung, “Branchynet: Fast inference via early exiting from deep neural networks,” in Proceedings of the 23rd International Conference on Pattern Recognition (ICPR). IEEE, 2016, pp. 2464–2469.

[19] S. Scardapane, M. Scarpiniti, E. Baccarelli, and A. Uncini, “Why should we share models? a survey on early-exit neural networks,” Cognitive Computation, vol. 12, no. 5, pp. 909–927, 2020.